

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2

Course Code:	DSA0404	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO5	Assessment:	Assessment Tool 2
Weightage:	40% of CIA	Total Marks:	100
CO3 Statement:	CO5-AT2 Coding Challenge		
Student Name:	T. Niranjan Kumar	Reg. No.:	192473007
Section / Batch:	CSE(DS)	Date:	26/08/26

Code of Conduct

I **T. Niranjan Kumar (192473007)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: **T. Niranjan**

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

PROGRAM:

```
import numpy as np

import matplotlib.pyplot as plt

from sklearn.metrics import roc_auc_score

# Actual class labels
y_true = np.array([0, 0, 0, 0, 1, 1, 1, 1])

# Predicted probabilities / scores
y_score = np.array([0.10, 0.20, 0.35, 0.40,
                    0.45, 0.60, 0.80, 0.90])

# Number of positive and negative samples
P = np.sum(y_true == 1)
N = np.sum(y_true == 0)

# Step 1: Sweep thresholds from high to low
thresholds = np.sort(np.unique(y_score))[:-1]

TPR = []
FPR = []

for threshold in thresholds:

    # Convert scores into predicted classes
    y_pred = (y_score >= threshold).astype(int)

    # Calculate TP and FP
    TP = np.sum((y_pred == 1) & (y_true == 1))
    FP = np.sum((y_pred == 1) & (y_true == 0))
```

```

# Calculate TPR and FPR manually

tpr = TP / P
fpr = FP / N

TPR.append(tpr)
FPR.append(fpr)

# Add starting and ending ROC points
FPR = np.array([0] + FPR + [1])
TPR = np.array([0] + TPR + [1])

# Step 2: Calculate AUC manually using trapezoidal rule
manual_auc = np.trapezoid(TPR, FPR)

# Step 3: Calculate AUC using sklearn
sklearn_auc = roc_auc_score(y_true, y_score)

# Display ROC points
print("ROC Curve Points:")
for i in range(len(FPR)):
    print(f'FPR = {FPR[i]:.2f}, TPR = {TPR[i]:.2f}')

print("\nManual AUC :", manual_auc)
print("Sklearn AUC:", sklearn_auc)

# Step 4: Validate
if np.isclose(manual_auc, sklearn_auc):
    print("Validation: Manual AUC matches sklearn AUC")

```

else:

```
print("Validation: AUC values do not match")
```

Step 5: Plot ROC curve

```
plt.plot(FPR, TPR, marker='o', label=f'AUC = {manual_auc:.2f}')
```

```
plt.plot([0, 1], [0, 1], linestyle='--')
```

```
plt.xlabel("False Positive Rate (FPR)")
```

```
plt.ylabel("True Positive Rate (TPR)")
```

```
plt.title("ROC Curve")
```

```
plt.legend()
```

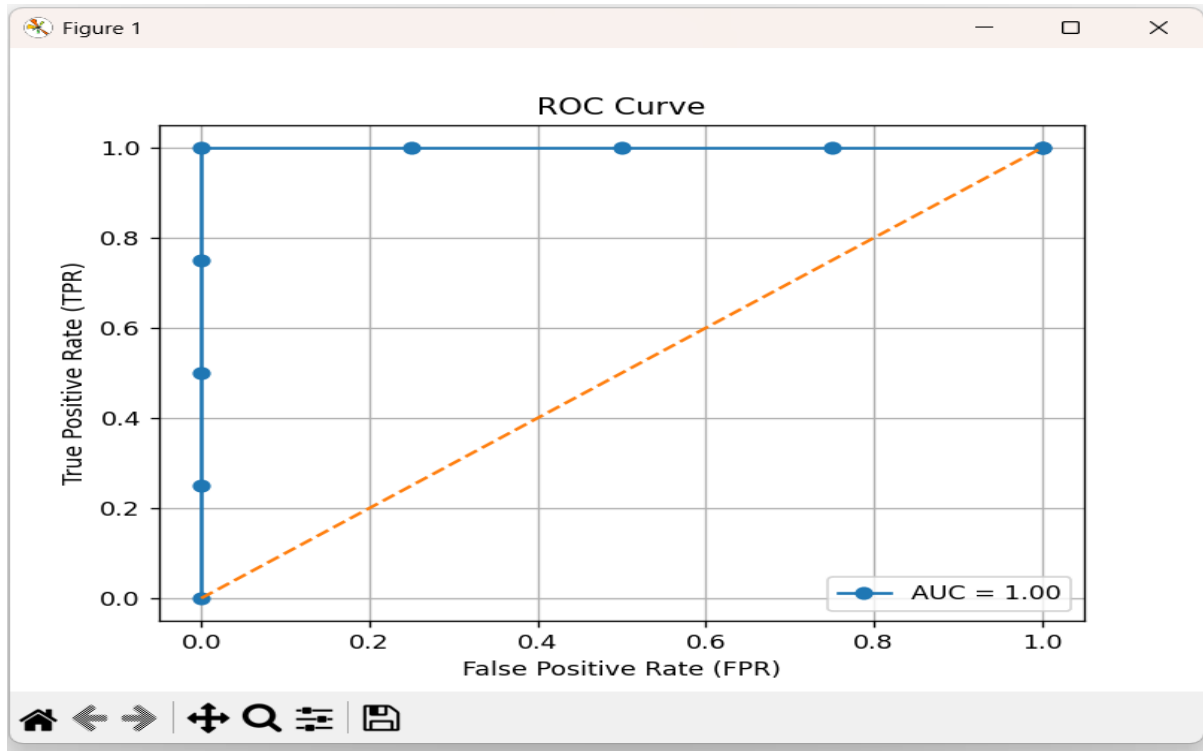
```
plt.grid()
```

```
plt.show()
```

OUTPUT:

```
*IDLE Shell 3.14.3*
File Edit Shell Debug Options Window Help
Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/thala/Downloads/FODS/CO5-AT2 Q.py =====
ROC Curve Points:
FPR = 0.00, TPR = 0.00
FPR = 0.00, TPR = 0.25
FPR = 0.00, TPR = 0.50
FPR = 0.00, TPR = 0.75
FPR = 0.00, TPR = 1.00
FPR = 0.25, TPR = 1.00
FPR = 0.50, TPR = 1.00
FPR = 0.75, TPR = 1.00
FPR = 1.00, TPR = 1.00
FPR = 1.00, TPR = 1.00

Manual AUC : 1.0
Sklearn AUC: 1.0
Validation: Manual AUC matches sklearn AUC
|
```



GEOTAG PHOTO:

